

YIDUO KE

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PUBLICATIONS AND WORKSHOP TALKS

An Algorithmic Approach to Address Course Enrollment Challenges *Biswas A., Ke Y., Khuller S., Liu Q. C. FORC 2023*

Scalable Auction Algorithms for Bipartite Maximum Matching Problems *Liu Q. C., Ke Y., Khuller S., AP-PROX 2023*

Fair Allocation of Conflicting Courses under Additive Utilities *Biswas A., Ke Y., Khuller S., Liu Q. C. AAMAS 2024, extended abstract*

EFX Allocation of Conflicting Courses under Additive Identical Utilities *Biswas A., Ke Y., Khuller S., Liu Q. C. KR Conventicle 2024 (UNSW)*

EDUCATION

Northwestern University *Ph.D. in Computer Science* *class of 2026*

Kolmogorov Complexity ◊ Modern Discrete Probability ◊ Mechanism Design ◊ Combinatorial Optimization ◊ Approximation Algorithms

Cornell University *B.S. in Computer Science (GPA: 3.82, Magna Cum Laude, Dean's List every semester)* *class of 2021*

Analysis of Algorithms ◊ Analysis of Boolean Functions ◊ Introduction to Computational Complexity ◊ Functional Programming in OCaml ◊ Machine Learning ◊ Computer Vision ◊ Artificial Intelligence ◊ Probability Models and Inference ◊ Linear Algebra ◊ Discrete Mathematics ◊ Object-Oriented Programming and Data Structures

WORK OR PROJECT EXPERIENCE

Software Engineering Intern *June 2021 - Sep 2021*
Raytheon *Riverdale, MD (remote)*

- worked in NASA's EOSDIS team
- deprecated ECHO-Token usage in NASA's Common Metadata Repository legacy services software

Software Engineering Intern *May 2020 - Aug 2020*
Raytheon *Riverdale, MD (remote)*

- worked in NASA's EOSDIS team
- wrote a user interface for NASA Earthdata using data from NASA's Common Metadata Repository
- wrote unit tests for my product to run against
- deployed my product using Atlassian Bamboo and AWS, thus going through the entire software development cycle

Software Engineering Intern *June 2020 - Aug 2020*
Pacific Northwest National Laboratory *Richland, WA (remote)*

- worked in a team of 4 to develop a virtual reality app in Unity from start to finish
- implemented and addressed dynamic client requirements in the app
- met with clients every 2 weeks to present updates on development progress and gain feedback

Research Intern *June 2024 - Sep 2024*
Espresso AI *Brooklyn, NY*

- developed offline scheduling algorithms for dbt batch queries to lower clients' Snowflake bills

TECHNICAL SKILLS

Languages	Python, Octave, OCaml, Java, C, C++, C#, Julia
Front-end skills	HTML/CSS/Bootstrap, JavaScript/jQuery/D3.js
Back-end frameworks	Flask, Node.js, Serverless, Unity
Cloud	AWS services
Databases	MySQL/SQLite/MongoDB
Tools	Git, Linux(Ubuntu), Atlassian Jira, Bamboo, BitBucket
Embedded Programming	RISC-V Assembly, reverse engineering